Revision of Grade 9 Algebra

Exercise 1

1. Multiply out and collect like terms where necessary:

a. $1-5(x+1)$ $1-5x-5=-5x-4$

b. $1-5(x-1)$ $1-5x+5=-5x+6$

c. $-2x(x+3)$ $-2x^2-6x$

d. $(x+3)(-2x)$ $-2x^2-6x$

e. $(x+3)-2x$ $x+3-2x=3-x$

f. $2x-2x(x+3)$ $2x-2x^2-6x$ $-2x^2-4x$

2. Multiply out and collect like terms where necessary:

a. $2n(n-4)-n(n-6)$ $2n^2-8n-n^2+6n$

b. $h(h-4)+2h(h-1)$ $h^2-4h+2h^2-2h$ $3h^2-6h$

c. $p(p+5)-5p(-p+4)$ p^2-2p

d. $x(x-3)-7x(x-2)-6x^2+11x$

e. $a(a-3)-a-4a^2$ $-3a^2-4a$

f. $5(t-2)(-3)+t(t+3)$ $t^2-12t+30$

g) $(m^2-n)(m^3+3n)$ $m^5+3m^2n-m^3n-3n^2$

h) $\left(3p+\frac{2}{3}\right)\left(2p-\frac{2}{5}\right)$ $6p^2-\frac{6}{5}p+\frac{4}{3}p-\frac{4}{15}$ $6p^2+\frac{2}{15}p-\frac{4}{15}$

i) $(2a^x-1)(4a^x+3)$ $8a^{2x}+6a^x-4a^x-3$ $8a^{2x}+2a^x-3$

j) $(a^{3x}-3b^{2y})(a^{3x}-3b^{2y})$ $a^{6x}-3a^{3x}b^{2y}-3a^{3x}b^{2y}+9b^{4y}$ $=a^{6x}-6a^{3x}b^{2y}+9b^{4y}$

https://www.geogebra.org/m/kzz6n2a9#material/kC9MDBHt

Exercise 2

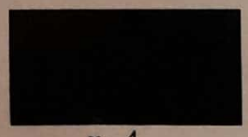
MAS Page 14 Exercise 1

Exercise 3

Find the area of the following quadrilaterals:

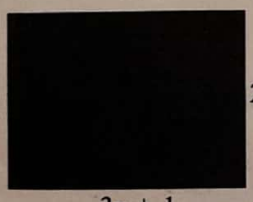
a. **Rectangles**

i)



x^2-4x

ii)

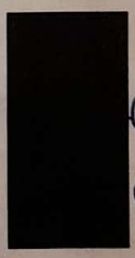


$(2x-4)(3x+1)$

$6x^2+2x-12x-4$

$6x^2-10x-4$

iii)



$(3x+y)(2x+7y)$

$6x^2+2xy+21xy+7y^2$

$6x^2+23xy+7y^2$

b. **Squares**

i)



$x^2-8x+16$

ii)



$\left(x-\frac{1}{x}\right)\left(x-\frac{1}{x}\right)$

$x^2-1-1+\frac{1}{x^2}$

$x^2-2+\frac{1}{x^2}$

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Exercise 4

Expand the following and simplify by adding like terms:

a. $(2a+1)(2a+1)$

b. $(m-1)(m-1)$ m^2-2m+1

c. $(n+3)^2$ n^2+6n+9

d. $(p+2)(p-2)$ p^2-4

e. $(w+4)^2$ $w^2+8w+16$

f. $(d+3)(d-3)$ d^2-9

g. $(3x-1)(3x-1)$ $9x^2-6x+1$

h. $(x+y)^2$ $x^2+2xy+y^2$

i. $(2x+1)^2$ $4x^2+4x+1$

j. $2(y+3)^2$ $2(y^2+6y+9)=2y^2+12y+18$

k. $-3(m+2)^2$ $-3(m^2+4m+4)$ $-3m^2-12m-12$

l. $-(n-3)^2$ $-(n^2-6n+9)$ $-n^2+6n-9$

The Product of a Binomial and a Trinomial :

1. $(x+1)(x^2-2x-5)$

$= x^3-2x^2-5x+x^2-2x-5$

$= x^3-x^2-7x-5$

2. $(a-3b)^3$

$= (a-3b)(a-3b)(a-3b)$

$= (a-3b)(a^2-6ab+9b^2)$

$= a^3-6a^2b+9ab^2-3a^2b+18ab^2-27b^3$

$= a^3-9a^2b+27ab^2-27b^3$

$=$

Exercise 5: MAS page 15 Exercise 2

Exercise 6

Expand the following and simplify by adding like terms:

a. $(m+1)(m^2-m+1)$ m^3+1

b. $(m-1)(m^2+m+1)$ m^3-1

c. $(n-2)(n^2+2n+4)$ n^3-8

d. $(p+2)(p^2-2p+4)$ p^3+8

e. $(d+3)(d^2-3d+9)$ d^3+27

f. $(d-3)(d^2+3d+9)$ d^3-27

g. $(x-4y)(x^2+4xy+16y^2)$ x^3-64y^3

h. $(2x-3y)(4x^2+6xy+9y^2)$ $8x^3-27y^3$

What do you notice about the answers in exercise 6 ? Make your own notes on these examples.

cube sum / cube difference

We will revisit these examples in the section on Factorising.

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Exercise 7: Consolidation of Grade 9 Products

1. If $A=(x+2)$ *and* $B=(x-1)$, find the value of :

a. A^2 b. B^2 c. AB d. $3A+2B$ e. AB^2

x^2+4x+6 x^2-2x+1 x^2+x-2 $3(x+2)+2(x-1)$ $3x+6+2x-2$ $5x+4$

$(x+2)(x^2-2x+1)$ $x^3-2x^2+x+2x^2-4x+2$ x^3-3x+2

2. If $W=(x+2)$ *and* $V=(x-1)$, find the value of:

a. W^2 b. V^2+W^2 c. $-2VW$ d. $3-2W$ e. W^2V

x^2+4x+4 $x^2-2x+1+x^2+4x+4$ $-2(x-1)(x+2)$ $-2(x^2+x-2)$ $-2x^2-2x+4$ $=2x^2+2x+5$

d) $3-2(x+2)$ $3-2x-4$ $-2x-1$

$(x-1)(x^2+6x+6)$ $x^3+4x^2+4x-x^2-4x-4$ x^3+3x^2-4

3. Expand by multiplying out and collect like terms:

a) $(x+2)^3$ $(x+2)(x^2+4x+4)$ $x^3+4x^2+4x+2x^2+8x+8$ $=x^3+6x^2+12x+8$

b. $(2x-3)^3$ $8x^3-36x^2+54x-27$

c. $(x-2)(x^2+2x+4)$ x^3-8

d. $(a+6)(a^2-6a+36)$ a^3+216

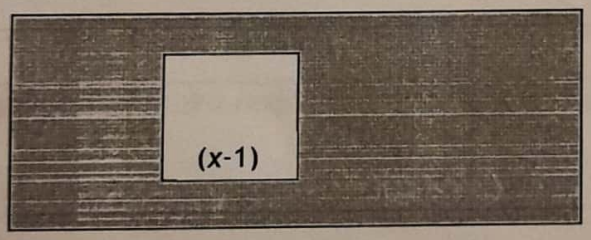
e. $-(x+y)^3$ $-(x+y)(x^2+2xy+y^2)$ $=-(x^3+2x^2y+xy^2+x^2y+2xy^2+y^3)$ $=-x^3-3x^2y-3xy^2-y^3$

f. $(9x^2+12xy+16y^2)(3x-4y)$ $27x^3-64y^3$

4. a. Find the area of a square with sides $(2m+3n)$. $4m^2+12mn+9n^2$

b. Find the area of a rectangle with sides $(2m+3n)$ and $(2m-3n)$. $4m^2-9n^2$

5. Find the area of the shaded area if the rectangle has sides of $(3x+1)$ and $(3-x)$ and the square has sides of $(x-1)$ metres long.



$(3x+1)(3-x)-(x-1)^2$ $9x-3x^2+3-x-(x^2-2x+1)$ $-4x^2+10x+2$

6. Expand: $(-3x^2+y)(x-4y)$, hence find the value of the expression if $x=-4$ and $y=3$

$-3x^3+12x^2y+xy-4y^2$ $=-3(-4)^3+12(-4)^2(3)+(-4)(3)-4(3)^2$ $=720$

7. If $(a+b)=7$ and $ab=-4$, determine the value of a^2+b^2.

$a^2+2ab+b^2=49 \rightarrow a^2+b^2+2(-4)=49$ $a^2+b^2=57$

8. If $\left(x+\frac{1}{x}\right)=6$ determine the value of:

a) $x^2+2+\frac{1}{x^2}=36$ $\rightarrow$ sq both sides $x^2+\frac{1}{x^2}=34$

a. $x^2+\frac{1}{x^2}$ 34

b. $x-\frac{1}{x}$ $\pm\sqrt{32}$

b) $x^2-2+\frac{1}{x^2}$ $=34-2=32$ $\rightarrow$ $\left(x-\frac{1}{x}\right)^2=32$ $x-\frac{1}{x}=\pm\sqrt{32}$

9. If $3x^2+8x+12=0$ determine the value of $3x^2+8x+23$.

$3x^2+8x=-12$ $-12+23$ $=11$

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